

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Description: Triethylamine trihydrofluoride, ca. 37% HF
Cat No. : C35062
Molecular Formula C6 H15 N.3HF

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

Details of the supplier of the safety data sheet
Importer

Fisher Scientific Korea
 D5,D6, Incheon Airport Logistics Complex
 150, Gonghangdong-Ro 296 Beon-Gil
 Jung-Gu, Incheon
 Tel: +82-1661-9555
 Fax: +82-2-2023-0603

Supplier

Thermo Fisher Scientific Chemicals, Inc.
 30 Bond Street
 Ward Hill, MA 01835-8099

E-mail address

Chem.KR@thermofisher.com

Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture
Physical hazards

Flammable liquids Category 4

Substances/mixtures corrosive to metal Category 1

Health hazards

Acute Oral Toxicity	Category 2
Acute Dermal Toxicity	Category 1
Acute Inhalation Toxicity - Vapors	Category 2
Skin Corrosion/Irritation	Category 1 A
Serious Eye Damage/Eye Irritation	Category 1

Environmental hazards

Based on available data, the classification criteria are not met

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

Label Elements



Signal Word

Danger

Hazard Statements

H227 - Combustible liquid

H290 - May be corrosive to metals

H300 + H310 + H330 - Fatal if swallowed, in contact with skin or if inhaled

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

Precautionary Statements

Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P234 - Keep only in original packaging

P264 - Wash hands and face thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P262 - Do not get in eyes, on skin, or on clothing

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P271 - Use only outdoors or in a well-ventilated area

P284 - Wear respiratory protection

Response

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction

P390 - Absorb spillage to prevent material damage

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P321 - Specific treatment (see supplemental first aid instructions on this label)

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P310 - Immediately call a POISON CENTER or doctor

P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P320 - Specific treatment is urgent (see supplemental first aid instructions on this label)

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P363 - Wash contaminated clothing before reuse

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Storage

P403 - Store in a well-ventilated place

P406 - Store in corrosion resistant polypropylene container with a resistant liner

P405 - Store locked up

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

Other Hazards

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

NFPA

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

Health
4

Flammability
2

Instability
1

Physical hazards
-

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	Common Name	CAS No	Index No	Weight %
Triethylammonium fluoride	No information available	73602-61-6	KE-05-1313	40 - 50
Water	Aqua	7732-18-5	KE-35400	20 - 30
Hydrogen fluoride	No information available	7664-39-3	KE-20198	20 - 30

SECTION 4: FIRST AID MEASURES

Description of first aid measures

General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

Inhalation

If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.

Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

Extinguishing media

Suitable Extinguishing Media

Carbon dioxide (CO₂). Powder. Water spray. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion. CO₂, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

Extinguishing media which must not be used for safety reasons

No information available.

Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Combustible material. Containers may explode when heated.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂), Nitrogen oxides (NO_x), Hydrogen fluoride.

Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition.

Conditions for Safe Storage, Including any Incompatibilities

Corrosives area. Store under an inert atmosphere. Protect from moisture. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

Specific End Uses

Use in laboratories.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS No	Korea	ACGIH TLV	OSHA PEL
Triethylammonium fluoride	73602-61-6	TWA: 2.5 mg/m ³	TWA: 2.5 mg/m ³	(Vacated) TWA: 2.5 mg/m ³
Water	7732-18-5	Not listed	Not listed	Not listed
Hydrogen fluoride	7664-39-3	Ceiling: 3 ppm TWA: 0.5 ppm TWA: 2.5 mg/m ³	TWA: 0.5 ppm TWA: 2.5 mg/m ³ Ceiling: 2 ppm	(Vacated) TWA: 3 ppm (Vacated) TWA: 2.5 mg/m ³ (Vacated) STEL: 6 ppm

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

		Skin	Skin	TWA: 3 ppm
Component	CAS No	European Union	The United Kingdom	Germany
Triethylammonium fluoride	73602-61-6	Not listed	Not listed	TWA: 1 mg/m ³ (8 Stunden). AGW - exposure factor 4 TWA: 1 mg/m ³ (8 Stunden). MAK Haut
Water	7732-18-5	Not listed	Not listed	Not listed
Hydrogen fluoride	7664-39-3	TWA: 1.8 ppm (8h) TWA: 1.5 mg/m ³ (8h) STEL: 3 ppm (15min) STEL: 2.5 mg/m ³ (15min)	STEL: 3 ppm 15 min STEL: 2.5 mg/m ³ 15 min TWA: 1.8 ppm 8 hr TWA: 1.5 mg/m ³ 8 hr	TWA: 1 ppm (8 Stunden). AGW - exposure factor 2 TWA: 0.83 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 1 mg/m ³ (8 Stunden). AGW - exposure factor 4 TWA: 1 ppm (8 Stunden). MAK TWA: 0.83 mg/m ³ (8 Stunden). MAK TWA: 1 mg/m ³ (8 Stunden). MAK Höhepunkt: 2 ppm Höhepunkt: 1.66 mg/m ³ Haut

ACGIH - Biological Exposure Indices

Component	CAS No	ACGIH - Biological Exposure Indices
Triethylammonium fluoride	73602-61-6	Not listed
Water	7732-18-5	Not listed
Hydrogen fluoride	7664-39-3	3 mg/g creatinine Medium: urine Time: prior to shift Determinant: Fluoride 10 mg/g creatinine Medium: urine Time: end of shift Determinant: Fluoride

Exposure Controls

Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles

Hand Protection

Protective gloves

Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Personal protective equipment

Use only those certified by the Korea Occupational Safety and Health Administration.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

Recommended Filter type: appropriate certified respirators
Multi-purpose/ABEK conforming to EN14387
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly
When RPE is used a face piece Fit Test should be conducted

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) Liquid

Odor pungent
Odor Threshold No data available
pH

Melting Point/Range -29 °C / -20.2 °F
Softening Point No data available
Boiling Point/Range 175 °C / 347 °F @ 15mmHg
Flash Point 87 °C / 188.6 °F **Method -** No information available

Evaporation Rate No data available
Flammability (solid,gas) Not applicable Liquid
Explosion Limits No data available

Vapor Pressure No data available
Vapor Density No data available (Air = 1.0)
Specific Gravity / Density 0.99 g/cm3 @ 20 °C
Bulk Density Not applicable Liquid
Water Solubility Miscible
Solubility in other solvents No information available

Partition Coefficient (n-octanol/water)

Component	CAS No	log Pow
Triethylammonium fluoride	73602-61-6	No data available
Water	7732-18-5	No data available
Hydrogen fluoride	7664-39-3	-1.4

Autoignition Temperature No data available
Decomposition Temperature No data available
Viscosity No data available
Explosive Properties explosive air/vapour mixtures possible
Oxidizing Properties No information available

Molecular Formula C6 H15 N.3HF
Molecular Weight 161.21

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

SECTION 10: STABILITY AND REACTIVITY

Reactivity

None known, based on information available

Chemical Stability

Hygroscopic.

Possibility of Hazardous Reactions

Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

Conditions to Avoid

Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition.

Incompatible Materials

Strong bases. Oxidizing agent.

Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO₂). Nitrogen oxides (NO_x). Hydrogen fluoride.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

Information on expected route of exposure

Inhalation Not an expected route of exposure.
Ingestion May be harmful if swallowed.
Eyes Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness.
Skin Avoid contact with skin. Causes burns. Skin Corrosion/Irritation. Harmful in contact with skin.

Information on Health Hazards

(a) acute toxicity;
Oral Category 2
Dermal Category 1
Inhalation Category 2

Toxicology data for the components

Component	CAS No	LD50 Oral	LD50 Dermal	LC50 Inhalation
Triethylammonium fluoride	73602-61-6	No data available	No data available	No data available
Water	7732-18-5	-	-	-
Hydrogen fluoride	7664-39-3	No data available	No data available	LC50 = 0.79 mg/L (Rat) 1 h

(b) skin corrosion/irritation; Category 1

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory No data available
Skin No data available

Component	CAS No	Test method	Test species	Study result
Triethylammonium fluoride	73602-61-6	No data available	No data available	No data available
Water	7732-18-5	No data available	No data available	No data available
Hydrogen fluoride	7664-39-3	No data available	No data available	No data available

(e) germ cell mutagenicity; No data available

Component	CAS No	Test method	Test species	Study result
Triethylammonium fluoride	73602-61-6	No data available	No data available	No data available
Water	7732-18-5	No data available	No data available	No data available
Hydrogen fluoride	7664-39-3	No data available	No data available	No data available

(f) carcinogenicity; No data available

Component	CAS No	Test method	Test species / Duration	Study result
Triethylammonium fluoride	73602-61-6	No data available	No data available	No data available
Water	7732-18-5	No data available	No data available	No data available
Hydrogen fluoride	7664-39-3	No data available	No data available	No data available

There are no known carcinogenic chemicals in this product

Component	CAS No	IARC	NTP	ACGIH	OSHA	UK
Triethylammonium fluoride	73602-61-6	Not listed	Not listed	Not listed	Not listed	Not listed
Water	7732-18-5	Not listed	Not listed	Not listed	Not listed	Not listed
Hydrogen fluoride	7664-39-3	Not listed	Not listed	Not listed	Not listed	Not listed

(g) reproductive toxicity; No data available

Component	CAS No	Test method	Test species / Duration	Study result
Triethylammonium fluoride	73602-61-6	No data available	No data available	No data available
Water	7732-18-5	No data available	No data available	No data available
Hydrogen fluoride	7664-39-3	No data available	No data available	No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

Other Adverse Effects

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

Component	CAS No	EU - Endocrine Disruptors Candidate List	EU - Endocrine Disruptors - Evaluated Substances	Japan - Endocrine Disruptor Information
Triethylammonium fluoride	73602-61-6	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

Hydrogen fluoride	7664-39-3	Not applicable	Not applicable	Not applicable
-------------------	-----------	----------------	----------------	----------------

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

Component	CAS No	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Triethylammonium fluoride	73602-61-6	No data available	No data available	No data available	No data available
Water	7732-18-5	No data available	No data available	No data available	No data available
Hydrogen fluoride	7664-39-3	LC50 = 660 mg/L, 48h (Leuciscus idus)	EC50 = 270 mg/L, 48h (Daphnia species)	No data available	No data available

Persistence and degradability

Persistence

Miscible with water, Persistence is unlikely, based on information available.

Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Hydrogen fluoride	-1.4	No data available

Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

Ozone Depletion Potential

Component	CAS No	Ozone Depletion Potential
Triethylammonium fluoride	73602-61-6	Not listed
Water	7732-18-5	Not listed
Hydrogen fluoride	7664-39-3	Not listed

Other adverse effects

No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport

UN-No	UN1790
Proper Shipping Name	HYDROFLUORIC ACID
Hazard Class	8
Subsidiary Hazard Class	6.1
Packing Group	II

IATA

UN-No	UN1790
-------	--------

ALFAAC35062

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

Proper Shipping Name HYDROFLUORIC ACID
Hazard Class 8
Subsidiary Hazard Class 6.1
Packing Group II

IMDG/IMO

UN-No UN1790
Proper Shipping Name HYDROFLUORIC ACID
Hazard Class 8
Subsidiary Hazard Class 6.1
Packing Group II
Marine Pollutant No hazards identified

Special Precautions for User No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

Component	CAS No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	ISHL	AICS
Triethylammonium fluoride	73602-61-6	KE-05-1313	X	277-550-5	-	-	X	X	-	X	-
Water	7732-18-5	KE-35400	X	231-791-2	X	X	-	X	X		X
Hydrogen fluoride	7664-39-3	KE-20198	X	231-634-8	X	X	-	X	X	X	X

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Triethylammonium fluoride	73602-61-6	Not applicable	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	Not applicable	Not applicable	Not applicable	Annex I - Y34

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Triethylammonium fluoride	73602-61-6	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Listed	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	Listed	Not applicable	Not applicable

Korean National Regulations

Component	CAS No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration
Triethylammonium fluoride	73602-61-6	Annex 1 - KE-05-1313	Not applicable	Not applicable
Water	7732-18-5	Annex 1 - KE-35400 Exempt (Index No. 25)	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	Annex 1 - KE-20198	Not applicable	Listed

Component	CAS No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
Triethylammonium fluoride	73602-61-6	2003-1-0540 (>25%)	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	1997-1-0382 (>1%)	Not applicable	Not applicable

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

Component	CAS No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Triethylammonium fluoride	73602-61-6	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	>1%	1000 kg/yr	150000 kg/yr

Component	CAS No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance
Triethylammonium fluoride	73602-61-6	> 25% (CCA)	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	> 1% (CCA) > 1% (ISHA)	Not applicable	Not applicable

CCA = Chemical Control Act

ISHA = Subject to Process Safety Reports

Component	CAS No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Triethylammonium fluoride	73602-61-6	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	Listed	Not applicable	Not applicable

Component	CAS No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Triethylammonium fluoride	73602-61-6	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	Listed	Listed	Not applicable

Component	CAS No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
Triethylammonium fluoride	73602-61-6	Not applicable	TWA: 2.5 mg/m ³	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	1000 kg 10000 kg	Ceiling: 3 ppm TWA: 0.5 ppm TWA: 2.5 mg/m ³ Skin	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
Triethylammonium fluoride	73602-61-6	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

NFA Dangerous Substances Classification

Class 4.5 (Group 3 petroleum, soluble) - 4000 L

Control Parameters

ALFAAC35062

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

Component	CAS No	Korea	ACGIH - Biological Exposure Indices
Triethylammonium fluoride	73602-61-6	TWA: 2.5 mg/m ³	Not listed
Water	7732-18-5	Not listed	Not listed
Hydrogen fluoride	7664-39-3	Ceiling: 3 ppm TWA: 0.5 ppm TWA: 2.5 mg/m ³ Skin	3 mg/g creatinine Medium: urine Time: prior to shift Determinant: Fluoride 10 mg/g creatinine Medium: urine Time: end of shift Determinant: Fluoride

US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

Component	CAS No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Triethylammonium fluoride	73602-61-6	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	Not applicable	TQ: 1000 lb

CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355)

Component	CAS No	CERCLA Extremely Hazardous Substances RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Triethylammonium fluoride	73602-61-6	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Hydrogen fluoride	7664-39-3	100 lb	100 lb	1.0 %

GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Danger.

H290 - May be corrosive to metals. H300 + H310 + H330 - Fatal if swallowed, in contact with skin or if inhaled. H314 - Causes severe skin burns and eye damage. Combustible liquid.

P280 - Wear protective gloves/protective clothing/eye protection/face protection. P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P310 - Immediately call a POISON CENTER or doctor/physician.

SECTION 16: OTHER INFORMATION

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

SAFETY DATA SHEET

Triethylamine trihydrofluoride, ca. 37% HF

Revision Date 12-Jun-2024

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Prepared By

Health, Safety and Environmental Department

Creation Date

15-Apr-2015

Revision Date

12-Jun-2024

Revision Number

3

Revision Summary

New emergency telephone response service provider.

MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet